Kaustubh Dua

Software developer · Electronics & Computer Engineering student · Class of 2029

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SUMMARY

Engineering student who builds full-stack and data tools end to end in Python and TypeScript: a React or Next.js front end, a FastAPI or Supabase back end, automated tests and CI. Projects state their limits plainly and make no unmeasured claims.

EDUCATION

Jaypee Institute of Information Technology (JIIT), Noida
B.Tech., Electronics & Computer Engineering (ECM). Expected graduation: 2029.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++17, SQL
Frameworks: React, Next.js, Vite, FastAPI, Streamlit, Tailwind CSS
Data and AI: PostgreSQL, PostGIS, Supabase, pandas, scikit-learn, Google OR-Tools, Ollama, Three.js and WebGL
Tools and testing: Git, GitHub Actions, pytest, Playwright, CodeQL, Docker, CTest

PROJECTS

NIRDHOOM: crop-residue coordination system

React 19, Vite, Supabase, PostgreSQL, PostGIS, OR-Tools, Leaflet

- Built a field-to-buyer workflow covering booking, operator jobs, dispatch, residue pooling and verification, as an installable PWA.
- Designed server-side API routes and Row Level Security so service keys never reach the browser; added SHA-256 evidence hashing and an offline upload queue.
- Integrated an OR-Tools dispatch service with a local fallback, and kept seeded demo data separate from live records. Status: prototype.

GitHub: kaustubhdua/nihdhoom Live demo: nihdhoom-9qiu.vercel.app

Saathi: voice-first multilingual assistant

Next.js, React, FastAPI, Python, speech APIs, GitHub Actions

- Implemented deterministic Hindi and Hinglish intent routing and checked it against a 120-case synthetic benchmark.
- Attached a source and retrieval time to every factual answer, and added consent prompts and human escalation.
- Set up CI for backend tests, front-end checks and dependency audits. Status: MVP.

GitHub: kaustubhdua/Saathi Live demo: saathi-two-pi.vercel.app

PatchPilot: pull-request risk triage

Python, FastAPI, pytest, Ruff, Docker, GitHub Actions

- Wrote a diff analyser that returns evidence-backed risk signals and a coarse 0-100 score without ever running the analysed code.
- Bounded untrusted input with size limits, strict schemas and rate limiting (HTTP 429 with Retry-After); shipped a non-root Docker image. Status: early stage.

GitHub: kaustubhdua/PatchPilot

Smart Waste Collection

Python, scikit-learn, OR-Tools, Streamlit, Folium

- Built a pipeline that predicts bin fill levels with a Random Forest, scores collection priority from 0 to 100 and plans truck routes with OR-Tools.
- Documented that the IoT-style data is synthetic and the routing is simulated. Status: prototype.

GitHub: kaustubhdua/smart_waste_project Live demo: smartwasteproject-glmvz89ttkxfktcssfdfuc.streamlit.app

JIIT Youth Club website

React 19, Vite, React Router, Supabase, Playwright, GitHub Actions

- Built the public site for the club (events, clubs, gallery, archive, recruitment) with Supabase-backed content.
- Added quality gates: static checks for security, SEO and accessibility, Playwright browser tests and CodeQL scanning.

GitHub: kaustubhdua/JYC-Website Live demo: jyc-website-eight.vercel.app

TEAM WORK AND OTHER PROJECTS

- **Adapt AI** (team project, contributor): local AI assistant that recommends an Ollama model for the machine's RAM and VRAM (FastAPI, Ollama). Upstream repository belongs to a teammate. My fork
- **This portfolio**: a scroll-driven 3D world with a hand-written WebGL 2 engine and no 3D library, covered by Playwright tests. Source
- **Explored (forks, not original work):** SATARK, a security-awareness triage workspace; GitGlobe, a 3D map of repositories. **Also built:** AkashChalak, Smart Bike Taxi (C++17), JIIT Toppers.